

Zikang Leng

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🏠 [zikangleng.github.io](https://github.com/zikangleng)

in [zikang-leng](https://github.com/zikang-leng)

🔗 [ZikangLeng](https://github.com/ZikangLeng)

🎓 [Google Scholar](https://scholar.google.com/citations?user=zikangleng)

EDUCATION

Georgia Institute of Technology

PhD in Computer Science

Atlanta, GA

Aug. 2024 - Expected May. 2028

- Advised by Prof. Thomas Plötz. Research areas: human activity recognition, data generation, LLMs, computer vision

Georgia Institute of Technology

B.S. in Computer Science (Theory & AI), B.S. in Physics | GPA: 4.0

Atlanta, GA

Aug. 2021 - May. 2024

EXPERIENCE

Graduate Research Assistant

Georgia Institute of Technology

Aug. 2024 - Present

Atlanta, GA

- Developed **AgentSense** (AAAI '26): LLM-guided agents simulated **18** personas across **22** home layouts to generate **250 days** of privacy-preserving smart-home sensor data; pretraining on it beat baselines on **5** real HAR datasets.
- Built **EduGage** (IMWUT '26), fusing **8** synchronized physiological, motion, and eye-tracking signals from 16 participants to reach **83.75%** within-1 engagement accuracy, outperforming foundation-model and LLM baselines.
- Benchmarked virtual IMU generation against classical augmentation over **100** activities and **22** body locations, across 3 HAR benchmarks and 4 model architectures.

Research Intern

BehaviorAI

Aug. 2026 – Dec. 2026

Brooklyn, NY

Research Intern, Sensor Machine Learning

Nokia Bell Labs

Jun. 2025 – Aug. 2025

Cambridge, United Kingdom

- Built a quality-aware fusion framework using ECG at training time to improve IMU- and PPG-only inference.
- Integrated self-supervised clustering and uncertainty estimation to improve robustness and interpretability.

Undergraduate Research Assistant

Georgia Institute of Technology

Aug. 2022 – May 2024

Atlanta, GA

- Developed **IMUGPT** (IMWUT '24), generating virtual IMU data from text via LLMs and motion synthesis; motion filtering and diversity metrics cut generation effort by **over 50%**.
- Evaluated alternative data-collection strategies for wheelchair transportation mode detection, reducing reliance on recruiting wheelchair users (Best Paper Nomination, ISWC '24).
- Built a personalized emotion-recognition system using wearable IMUs and virtual motion data for pretraining.
- Built **FingerSpeller** (ASSETS '23), a camera-free ASL fingerspelling system reaching **91%** character accuracy from smart rings; released a 42,288-letter dataset.

Machine Learning Research Intern – SULI

Argonne National Laboratory

May 2023 – Aug. 2023

Lemont, IL

- Accelerated GNN training for ocean simulation by **213×** using distributed training across **256 GPUs**.
- Expanded GNN training data via inverse-distance-weighted interpolation for larger-scale ocean simulation.

SELECTED PUBLICATIONS

EduGage: Methods and Dataset for Sensor-Based Momentary Assessment of Engagement in Self-Guided Video Learning UbiComp/IMWUT '26 [\[arxiv\]](#)

Zikang Leng*, Edan Eyal*, Yingtian Shi, Jiaman He, Yaqi Liu, Thomas Plötz

AgentSense: Virtual Sensor Data Generation Using LLM Agents in Simulated Home Environments AAAI '26 [\[paper\]](#)[\[code\]](#)

Zikang Leng*, Megha Thukral*, Yaqi Liu*, Hrudhai Rajasekhar, Shruthi K Hiremath, Jiaman He, Thomas Plötz

Can We Hide Machines in the Crowd? Quantifying Equivalence in LLM-in-the-loop Annotation Tasks SIGIR-AP '25 [\[paper\]](#)[\[code\]](#)

Jiaman He, Zikang Leng, Dana McKay, Damiano Spina, Johanne R Trippas

Characterising Topic Familiarity and Query Specificity Using Eye-Tracking Data SIGIR '25 [\[paper\]](#)[\[code\]](#)

Jiaman He, Zikang Leng, Dana McKay, Johanne R Trippas, Damiano Spina

*Equal Contribution

Scaling Human Activity Recognition: A Comparative Evaluation of Synthetic Data Generation and Augmentation Techniques <i>Zikang Leng, Archith Iyer, Thomas Plötz</i>	arXiv '25 [arxiv]
IMUGPT 2.0: Language-Based Cross Modality Transfer for Sensor-Based Human Activity Recognition <i>Zikang Leng, Amitrajit Bhattacharjee, Hrudhai Rajasekhar, Lizhe Zhang, Elizabeth Bruda, Hyeokhyen Kwon, Thomas Plötz</i>	UbiComp/IMWUT '24 [paper][code]
We Need More Data for People with Disabilities: A Comparative Study on Data Collection for Wheelchair Transportation Mode Detection (Best Paper Nomination) <i>Sungjin Hwang*, Zikang Leng*, Seungwoo Oh, Kwanguk Kim, Thomas Plötz</i>	UbiComp/ISWC '24 [paper]
Emotion Recognition on the Go: Utilizing Wearable IMUs for Personalized Emotion Recognition <i>Zikang Leng*, Myeongul Jung*, Sungjin Hwang, Seungwoo Oh, Lizhe Zhang, Thomas Plötz, Kwanguk Kim</i>	UbiComp'24 HASCA [paper]
FingerSpeller: Camera-Free Text Entry Using Smart Rings for American Sign Language Fingerspelling Recognition <i>David Martin*, Zikang Leng*, Tan Gemicioglu, Jon Womack, Jocelyn Heath, Bill Neubauer, Hyeokhyen Kwon, Thomas Plötz, Thad Starner</i>	ASSETS '23 [paper]
Generating Virtual On-body Accelerometer Data from Virtual Textual Descriptions for Human Activity Recognition (Best Paper Honorable Mention) <i>Zikang Leng, Hyeokhyen Kwon, Thomas Plötz</i>	UbiComp/ISWC '23 [paper] [code] [news]
On the Utility of Virtual On-body Acceleration Data for Fine-grained Human Activity Recognition <i>Zikang Leng, Yash Jain, Hyeokhyen Kwon, Thomas Plötz</i>	UbiComp/ISWC '23 [paper]
PATENT	
Language-based Cross-modality Transfer System for Generating Virtual Inertial Measurement Unit Data to Enhance Human Activity Recognition	US Patent Filed
A System for Quality-Aware Multimodal Fusion Representation Learning with Uncertainty Estimation for Health Condition Modelling	US Patent Filed
AWARDS	
NSF Graduate Research Fellowship (GRFP) • Three-year \$37,000 annual stipend plus \$16,000 yearly tuition allowance for outstanding STEM graduate students	2024
Georgia Tech President's Fellowship • Awarded to select doctoral students for exemplary scholarship and innovation. \$5,500 annual stipend, renewable for three years	2024
Georgia Tech Institute Fellowship • Georgia Tech's most prestigious graduate fellowship, awarded to a select few President's Fellowship holders. \$2,000 annual stipend	2024
Provost's Academic Excellence Award • Top graduating senior from Georgia Tech's College of Computing. \$2,000 award	2024
President's Undergraduate Research Awards (Salary and Travel) • \$2,500 to conduct undergraduate research with Georgia Tech faculty, plus a travel grant	2023
TECHNICAL SKILLS	
Languages: Python, C/C++, Java, Bash, LaTeX	
ML & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, NumPy, OpenCV, Ray Tune	
Infrastructure: Distributed multi-GPU training, CUDA, Docker, Git	
MENTORED STUDENTS	
5 masters and 5 undergraduate students: Amitrajit Bhattacharjee, Elizabeth Bruda, Jocelyn Heath, Archith Iyer, Yaqi Liu, Ruijia Peng, Hrudhai Rajasekhar, Kalyan Salkar, Zijun Wang, Lizhe Zhang	